

Sieve of Eratosthenes & The Calculation of π

This document details the synergy between the Sieve of Eratosthenes, memory-efficient logistics, and the estimation of π using Euler's product formula.

1. The Sieve of Eratosthenes

The Sieve of Eratosthenes is a classic algorithm used to isolate prime numbers up to a limit n . By iteratively marking multiples of each prime, we avoid the heavy computation of trial division.

2. Logistical Optimization: Segmented Sieving

To compute primes for very large n , storing a complete bit-array leads to significant memory overhead. A **segmented sieve** approach solves this by dividing the range $[2, n]$ into smaller chunks (segments) of size Δ . This ensures the memory footprint remains constant regardless of the total limit.

3. Euler's Product Formula

The calculation of π can be derived from the distribution of prime numbers. Euler demonstrated that the Riemann Zeta function at $s=2$ is equal to an infinite product over all prime numbers p :

$$\zeta(2) = \sum (1/n^2) = \prod (1 - p^{-2})^{-1} = \pi^2/6$$

Rearranging for π gives:

$$\pi = \sqrt{6 / \prod (1 - p^{-2})^{-1}}$$

4. Implementation Approach

The following logic structure maintains efficiency:

- Precompute small primes up to $\sqrt{n}$.
- Segment the range $[2, n]$ into blocks.

3. For each block, sieve using small primes.
4. Update the running product of $(1 - 1/p^2)^{-1}$.
5. Finalize estimation using the square root of $(6 * \text{product})$.